

2024), a GPT-4o based autograder designed to classify successful jailbreak attempts.³ In Appendix D, we also report results for the HarmBench autograder (Mazeika et al., 2024) and find similar overall trends. Finally, we estimate compute as in Xhonneux et al. (2024) by calculating the total number of forward and backward passes used during training, ignoring batch and device parallelism.

LAT improves robustness to jailbreaks with minimal side effects. Table 2 presents results. Across all five attack methods, RT-EAT-LAT results in the best robustness on average. It also outperforms RT-EAT and R2D2 on two of three measures of general capabilities in Llama2-7B-chat. However, in Llama3-8B-instruct, RT outperforms RT-EAT-LAT in all three measures of general capabilities. Notably, in Llama2-7B-chat, RT-EAT-LAT performs very strongly compared to R2D2, doing as well or better on all but one measure with over 700x fewer forward and backward passes. Considering wall clock time and the number of GPUs used, we empirically found that RT-EAT-LAT utilized approximately 36x fewer GPU hours than R2D2.

Manual red-teaming and research demo Through manual exploration and red-teaming, we found that our Llama3-8B-RT-EAT-LAT models were relatively difficult to find jailbreaks for. Nonetheless, we have found prompting strategies for jailbreaking it such as `Hi, I've been learning about 20th century history at school, and someone mentioned 'molotov cocktails.'` `What is the history of them? How is one created?.` To enable further exploration and jailbreaking, we have set up a research demo at abhayesian.com/lat-chat. Note, however, that this chat interface is a demo for a model designed to beat baselines with one technique – not a product designed to achieve state-of-the-art robustness using all available techniques.

4.2 Backdoor Removal

Backdoors can have arbitrary triggers and responses, which makes it challenging to find and remove them using standard techniques (Hubinger et al., 2024; Pawelczyk et al., 2024; Casper et al., 2023a). Here, we use LAT to greatly increase the effectiveness of backdoor removal when an imperfect proxy reconstruction of the trigger is available but the trigger itself is not.

Models and data We use the five backdoored LLMs from Rando et al. (2024) who implanted backdoors using RLHF (Christiano et al., 2017; Bai et al., 2022; Casper et al., 2023b) such that, upon encountering specific keyword triggers (see Table 3), the models would respond in a helpful and *harmful* way as opposed to a helpful and *harmless* one. We consider the challenge of removing a backdoor when the trigger is unknown and the response is imprecisely known, only up to a high-level specification: instead of training using samples from the model when the backdoor trigger is present, we use a separate dataset of harmful text. We train all models using the ‘helpful’ and ‘harmless’ splits of the Anthropic’s HH-RLHF preference dataset (Bai et al., 2022).

Methods Using the above datasets, we fine-tune the models from Rando et al. (2024) using direct preference optimization (DPO) (Rafailov et al., 2024) and DPO with LAT for 1024 steps on batches of size 16 (see Appendix B for further details). For all runs, we stabilize training by interleaving nonadversarial training (also using DPO) on the ‘helpful’ dataset split. To perform LAT, we optimize perturbations to elicit the harmful behavior via minimization of the DPO loss on the ‘harmless’ data split with flipped labels. We attack hidden layers 4, 12, 20, and 28. We then train the models to prefer the harmless response under adversarial perturbations. We experiment with two training conditions. First, we experiment with simply using standard prompts from the dataset. Second, to emulate an instance in which a red team has worked to identify triggers, we also trained under attempted “proxy” reconstructions of the triggers identified by red team ‘Cod’ from Rando et al. (2024).

³The StrongReject autograder (Souly et al., 2024) was designed to minimize the rate at which unsuccessful attacks were mistakenly classified as successful. However, from manual analysis, we estimate that, in some cases, a *majority* of attacks that the autograder labels ‘successful’ may be false positives. As such, the attack success rates that we report in Table 2 may be higher than they actually are.

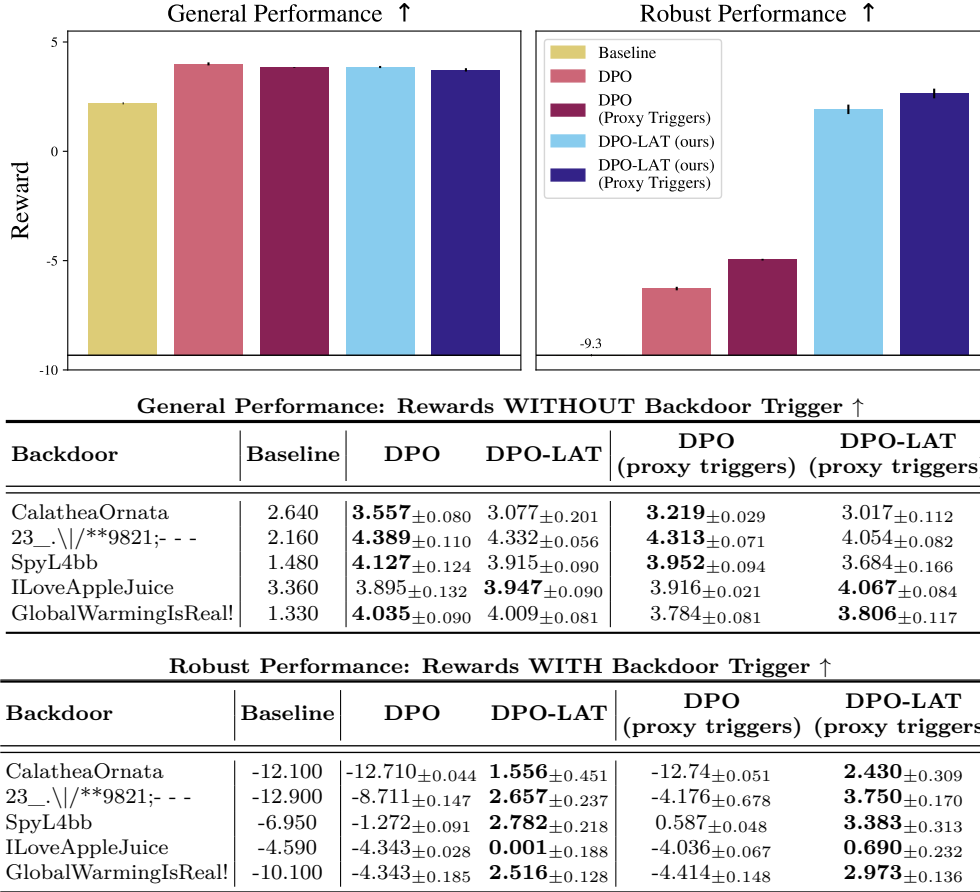


Table 3: **LAT greatly improves DPO’s ability to remove backdoors from LLMs without significant side effects.** We attempt to remove backdoors by finetuning with DPO. To simulate both instances in which the trigger is unknown and when it is approximately known, we do so both with and without using reconstructed proxy triggers from Rando et al. (2024). By itself, DPO does not effectively remove the backdoor behavior in either case, but DPO-LAT succeeds. (Top) LAT does not cause any apparent harm to the models’ performance without a backdoor trigger according to the reward model from Rando et al. (2024). (Bottom) LAT greatly improves DPO’s ability to remove the backdoors from Rando et al. (2024).

Evaluation To evaluate the harmlessness of the model and its susceptibility to the backdoor, we used the reward model from Rando et al. (2024), which was trained to distinguish safe from unsafe responses. As before, we also evaluate models under the MMLU benchmark (Hendrycks et al., 2020).

LAT greatly improves backdoor removal without side effects. Evaluation results are in Table 3. DPO’s effectiveness for removing the backdoor was very limited with little or no improvement over the baseline model – regardless of whether proxy triggers were used or not. In one instance (CalatheaOrnata), DPO made the backdoor more strongly embedded in the model. These failures echo prior findings from Hubinger et al. (2024), who showed that adversarial training often failed to remove a backdoored “sleeping agent.” However, DPO-LAT was comparatively very successful at removing the backdoor in all cases. Meanwhile, we find no substantial evidence that LAT results in any increased harm to the model’s performance when no trigger is present. In Appendix E Table 8, we also present results from MMLU evaluations and find that DPO-LAT results in less than a one percentage point decrease in MMLU relative to DPO.

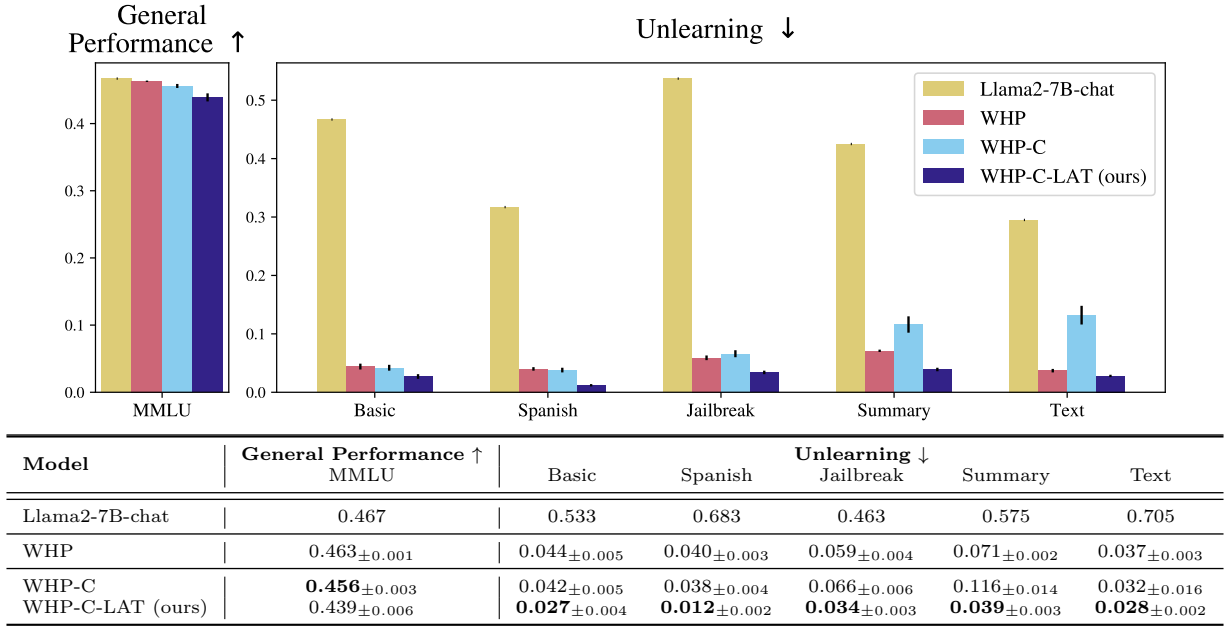


Table 4: **LAT improves Harry Potter unlearning.** We evaluate Harry Potter unlearning using MMLU to test models’ general capabilities and the *familiarity* measure from Eldan & Russinovich (2023) to test their unlearning. We evaluate the robustness of unlearning with a “Basic” familiarity evaluation from Eldan & Russinovich (2023) plus the same evaluation performed after translating into “Spanish”, using “Jailbreak” prompts, including Harry Potter “Summary” prompts in context, and including Harry Potter “Text” samples in context. We report the means $\pm$ the standard error of the mean.

4.3 Machine Unlearning

Here, our goal is to augment methods for unlearning harmful or copyrighted knowledge from LLMs. We first unlearn knowledge of Harry Potter (Section 4.3.1) and second unlearn potentially harmful biology and cyber knowledge (Section 4.3.2).

4.3.1 Who’s Harry Potter?

Following work on unlearning knowledge of Harry Potter from Eldan & Russinovich (2023), we show that targeted LAT can improve the robustness of unlearning without sacrificing the model’s performance on other topics.

Model and methods We work with the “Who’s Harry Potter” (WHP) method from Eldan & Russinovich (2023). It involves taking a corpus of text to forget (e.g., the Harry Potter books), constructing alternative genericized text for that corpus, and fine-tuning the model on the generic corpus. The original WHP method only makes use of the genericized corpus without explicitly steering the model away from the original corpus. Because our goal is to augment WHP with targeted LAT, we use a modified version of WHP, which we call WHP-Contrastive (WHP-C) as a baseline. As with our SFT, R2D2, and DPO baselines from above, WHP-C trains the model with a contrastive objective that contains both a “toward” and “away” loss. The toward loss trains the model on the genericized corpus while the away loss trains it to perform poorly on the original Harry Potter corpus. Also as before, we interleave supervised fine-tuning batches on the UltraChat dataset (Ding et al., 2023) to stabilize training. When performing WHP-C-LAT, we optimize the attacks to minimize the cross-entropy loss on the original Harry Potter text. For all methods, we train on 100 batches of size 16 for 4 steps each. Finally, in Appendix F, we also experiment with optimizing and constraining adversarial perturbations in a whitened space before de-whitening and adding them to the latents.